THE INTERPLAY BETWEEN INTERACTIONS AND TOPOLOGY/GEOMETRY IN CONDENSED MATTER PHYSICS

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Abstract

This thesis aims to introduce methods of quantum field theory for many-body systems and to apply them in the study of the impact of electron-electron interactions on the Hall conductivity of massive Dirac fermions, in a hexagonal lattice with a "special" gap. The Hall conductivity is determined from the transverse current-current response function in the limit of zero frequency, using the Kubo formula. When the Fermi energy is inside the gap, the conductivity of a system that breaks time-reversal symmetry is $\sigma_H = -e^2/h$, which is obtained by adding the contributions from both the K and K' valleys. Connecting this to the Berry curvature of the band structure, the Chern number is found to be $C_1 = -1$. Switching from the response to the time-ordered correlation function, the latter is expanded into an infinite power series of the coupling constant. Using diagrammatic perturbation theory, the correlation function is expressed in terms of the sum of the products of Feynman propagators, which at zero-order yield the same Hall conductivity as the linear response function approach. Trying to evaluate the first-order correction, both infrared and ultraviolet divergences are encountered, and therefore possible methods of regularisation are discussed. Despite the fact that the proposed correction proves to be insufficient for strong interactions, this formalism paves the way for further introducing "building blocks" to describe infinite-order contributions, which can be evaluated by means of integral equations.

1 Introduction

The study of many-body systems is one of the most complex and challenging fields in theoretical physics, and in particular in condensed matter. Since it requires the understanding of the behaviour of a macroscopic number of interacting variables, problems of this kind are very specific and hard to describe theoretically (Kotov et al., 2012). In this thesis, we focus on the impact of electron-electron interaction on the Hall conductivity of two-dimensional massive Dirac fermions in graphene, a field of strong experimental and theoretical interest (Neto et al., 2009; Kozikov et al., 2010). Our approach employs quantum field theory methods, where the Hall conductivity is related to the transverse current response function and the interactions are introduced via Feynman diagrams.

Interactions between particles give rise to a plethora of fascinating physical phenomena in many-body systems, from the band structure of crystals and superconductivity in metals (Bardeen et al., 1957), to the fractionalisation of the electron charge in the fractional quantum Hall effect (Stormer et al., 1999). Recent studies show that graphene hosts a unique electron system from the point of view of interactions, which play an important role in the low-temperature conductivity near the K point (Kozikov et al., 2010; Neto et al., 2009). This motivates us to further study the effects of these interactions on the conductivity of Dirac fermions, by means of diagrammatic perturbation theory.

After the discovery of the quantum Hall effect, most of the theoretical effort was directed at the quantisation of the conductivity, which can be related to a topological invariant without taking into account electron interactions (Thouless et al., 1982). However, this raises the question whether the Hall conductivity is robust to the effects of interactions (Zhang, Zubkov, 2020), something we aim to investigate for the particular case of massive Dirac fermions. In addition to this, we are also interested in their impact on the non-topological part of the conductivity, which is solely due to the doping of the material.

This semester's thesis focuses on bridging the gap between topology and the quantisation of the Hall conductivity, introducing the basis of linear response theory and diagrammatic perturbative calculations, while also trying to evaluate the first-order correction to the correlation function, in the zero-temperature formalism. Projected into the next semester's agenda, we plan to use renormalisation methods in order to remove the divergences we encounter, to extend this formalism for finite temperatures, and to ultimately go beyond first order in an attempt to generate non-perturbative results from an infinite-order resummation of the interaction power series.

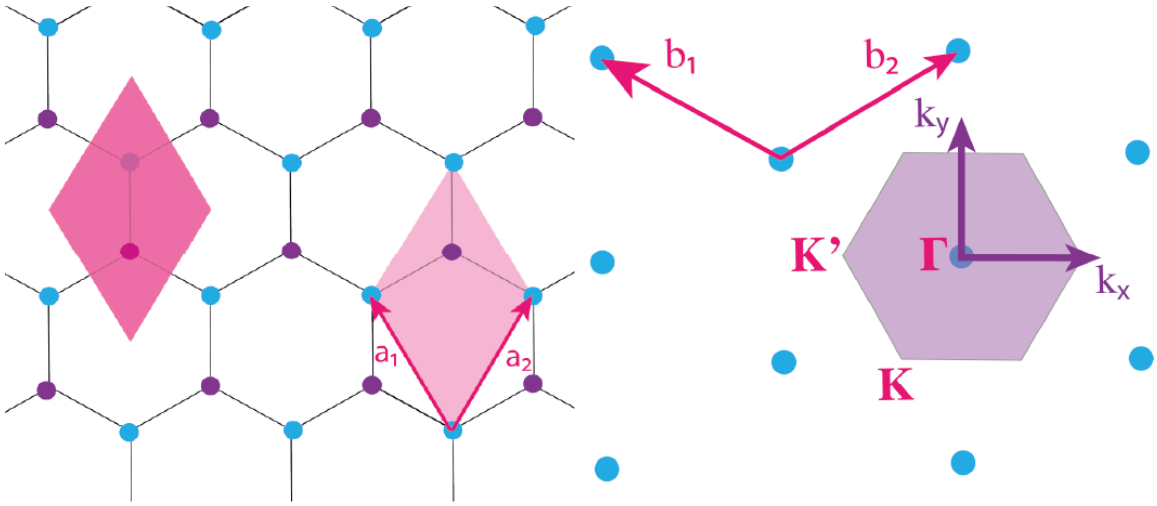
2 Topological and Geometrical Aspect

2.1 Topology

Topology has become increasingly prevalent in condensed matter these days, and despite its initial unexpected connection to physics, it has provided the framework to understand and predict a variety of novel physical phenomena. Topology is the abstract study of continuity and it focuses on the features of structures that remain the same when continuous changes are made to the parameters of the system. These continuous deformations can

be made through stretching, twisting, crumpling and bending, but not tearing or gluing (Bhattacharjee et al., 2017).

Accordingly, when searching for certain aspects that remain invariant under continuous deformations, this proves to be a vital tool for physicists. Most of the time these features appear when imposing boundary conditions since those determine the geometry and subsequently the topology of the space. As an example, one can consider the first Brillouin zone of graphene (or hexagonal boron nitride), illustrated in Figure 2.1. Imposing the periodic boundary conditions on this Brillouin zone corresponds to deforming the two-dimensional space into a torus, by matching the sides of the hexagon which are periodic. A step by step illustration of this deformation can be seen in Figure 2.2.



(a) Real space lattice (b) Reciprocal space lattice

Figure 2.1: Lattice structures of graphene (for the case of two identical carbon atoms in the unit cell) or of hexagonal boron nitride (alternating boron and nitrogen atoms). The shaded region in (b) is the first Brillouin zone, obtained using the Wigner-Seitz construction method. It is important to note the alternating K and K' points on its vertices.

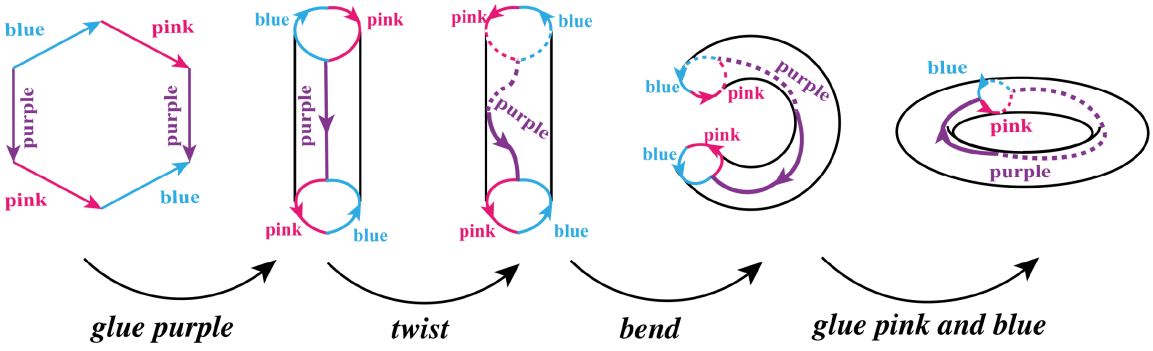


Figure 2.2: Continuous deformation of the first Brillouin zone of a hexagonal lattice. The periodic boundary conditions are imposed on the coloured arrows (pink, blue and purple).

2.2 Berry Phase, Connection and Curvature

Before one dives into the topological aspects of condensed matter physics, the key concept of Berry phase needs to be introduced. We consider a generic Hamiltonian which depends on time through the set of parameters $\boldsymbol{\mathcal{R}}(t) = (\mathcal{R}_1(t), \mathcal{R}_2(t), ...\mathcal{R}_N(t))$, and we are interested in its adiabatic evolution as $\boldsymbol{\mathcal{R}}(t)$ "moves" slowly along a path $\mathcal{C}$ in parameter space. For an instantaneous orthonormal basis of the Hamiltonian eigenstates, we can write:

$$\hat{H}[\boldsymbol{\mathcal{R}}(t)]\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle = \varepsilon_n[\boldsymbol{\mathcal{R}}(t)]\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle\,, \tag{2.1}$$

which can only determine the basis $|n[\boldsymbol{\mathcal{R}}(t)]\rangle$ up to a phase factor.

Note that the adiabatic nature of the process imposes that a system initially in one of its eigenstates $|\psi_n(t=0)\rangle = |n[\boldsymbol{\mathcal{R}}(t=0)]\rangle$ remains in that eigenstate throughout the entire process, while accumulating a phase (Griffiths, Schroeter, 2018):

$$|\psi_n(t)\rangle = C_n(t)\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle\,. \tag{2.2}$$

The arbitrariness of Equation (2.1) can be removed by making a phase choice, also known as a gauge, such that Equation (2.2) becomes:

$$|\psi_n(t)\rangle = e^{i\gamma_n(t)} \exp\left[-i\int_0^t dt' \varepsilon_n\left[\boldsymbol{\mathcal{R}}\left(t'\right)\right]\right] |n[\boldsymbol{\mathcal{R}}(t)]\rangle\,. \tag{2.3}$$

Here and throughout the text we take $\hbar = 1$. The second exponential is known as the dynamical phase factor, while $\gamma_n(t)$ is the Berry phase and accounts for a special aspect of the state evolution which is not captured by the second factor (Xiao et al., 2010).

Plugging our state into the Schrödinger Equation $i\partial_t|\psi_n(t)\rangle = \hat{H}[\boldsymbol{\mathcal{R}}(t)]|\psi_n(t)\rangle$ and taking the inner product with $\langle\psi_n(t)|$ on both sides, we get an expression for the Berry phase:

$$\frac{\partial}{\partial t}\gamma_n(t) = i\,\langle n[\boldsymbol{\mathcal{R}}(t)]|\,\frac{\partial}{\partial t}\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle\,. \tag{2.4}$$

When evaluating the time derivative of the basis eigenvectors, one can write the Berry phase in terms of a new quantity $\boldsymbol{\mathcal{A}}^n(\boldsymbol{\mathcal{R}})$[1], called the Berry connection or the Berry vector potential. The steps followed are shown here:

$$\frac{\partial}{\partial t}\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle = \sum_{i=1}^{N} \underbrace{\Big(\frac{\partial}{\partial \mathcal{R}_i}\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle\Big)}_{\nabla_{\boldsymbol{\mathcal{R}}}(...)\cdot\boldsymbol{e}_i} \frac{d\mathcal{R}_i}{dt} = \nabla_{\boldsymbol{\mathcal{R}}}\,|n[\boldsymbol{\mathcal{R}}(t)]\rangle \cdot \frac{d\boldsymbol{\mathcal{R}}}{dt}, \tag{2.5a}$$

$$\gamma_n(t) = \int_{\mathcal{C}} d\boldsymbol{\mathcal{R}} \cdot i\,\langle n(\boldsymbol{\mathcal{R}})|\,\nabla_{\boldsymbol{\mathcal{R}}}\,|n(\boldsymbol{\mathcal{R}})\rangle = \int_{\mathcal{C}} d\boldsymbol{\mathcal{R}} \cdot \boldsymbol{\mathcal{A}}^n(\boldsymbol{\mathcal{R}}). \tag{2.5b}$$

However, from the way we defined Berry connection it can clearly be seen that it is gauge-dependent, meaning that we can always find a suitable choice of gauge to cancel out the phase γ_n accumulated on any open path $\mathcal{C}$. This phase was neglected in the theoretical treatment of time-dependent problems for the most part until Michael Berry (Berry, 1984) first considered an adiabatic process where the motion was over a closed loop in parameter space. In this case he showed that the gauge-dependence disappears, making Berry phase

[1]Here we used the index n as a superscript because the quantity it is accompanying is a vector, and therefore it requires additional indices which are to be written as subscripts.

observable and having physical consequences (when it is non-vanishing). Also notable is the fact that γ_n does not depend explicitly on time, but rather on the path that $\mathcal{R}$ traverses, which emphasises the geometric origin of the phase (Girvin, Yang, 2019).

The analogy with electrodynamics should be fairly evident: $\boldsymbol{\mathcal{A}}^n \rightarrow$ vector potential of a magnetic field and $\gamma_n \rightarrow$ magnetic flux passing through a loop (in parameter space). For this analogy to be complete, the last concept we need to introduce is the Berry curvature Ω^{n1}. In three dimensions, it corresponds to the magnetic field from the Berry vector potential[2] - Equation (2.6a), while for the special case of a two-dimensional parameter space, it takes the form given in Equation (2.6b) (Vanderbilt, 2018):

$$\boldsymbol{\Omega}^n(\boldsymbol{\mathcal{R}}) = \nabla_{\boldsymbol{\mathcal{R}}} \times \boldsymbol{\mathcal{A}}^n(\boldsymbol{\mathcal{R}}), \tag{2.6a}$$

$$\Omega^n = \partial_x \mathcal{A}_y - \partial_y \mathcal{A}_x = -2\,\mathrm{Im}[\langle \partial_x n | \partial_y n \rangle]. \tag{2.6b}$$

2.3 Chern Theorem

The previous sections are linked through Chern's theorem, which states that the integral of the Berry curvature over a closed surface is quantised in units of 2π. For concreteness, we consider $\mathcal{M}$ to be a two-dimensional closed surface that can be embedded in three dimensions - like our Brillouin zone with periodic boundary conditions given as an example in Section 2.1. In mathematical terms, the theorem can be written as:

$$\int_{\mathcal{M}} d\boldsymbol{S} \cdot \boldsymbol{\Omega}^n(\boldsymbol{\mathcal{R}}) = 2\pi C_1, \tag{2.7}$$

where $C_1 \in \mathbb{Z}$ is called a Chern number of the first class (Resta, Vanderbilt, 2015).

We must emphasise that the Chern number is a topological invariant of the wave-function, and therefore going from one number to another would correspond to a phase transition or a quantum critical point (Bhattacharjee et al., 2017).

The Chern number made its first appearance in condensed matter physics in the famous TKNN paper on the quantum Hall effect (Thouless et al., 1982), and later led to major breakthroughs in the field, which were awarded the 2016 Nobel prize.

2.4 Geometry

The Chern theorem is a non-trivial generalisation of the classic Gauss-Bonnet theorem, which bridges the gap between topology and differential geometry. Its form is given by:

$$\frac{1}{2\pi} \int_{\mathcal{M}} dS\, K = \chi(\mathcal{M}) = 2(1-g), \tag{2.8}$$

where $\mathcal{M}$ is a closed smooth surface, K is the Gaussian curvature (equal to the product of the principal curvatures at any the given point), χ is the Euler characteristic (a topological invariant which describes the structure of the manifold regardless of the way it is bent), and g is the genus (it counts the number of handles of that manifold).

[2] In more than three dimensions, it is defined as a field tensor: $\Omega^n_{\mu\nu} = \partial_{\mathcal{R}_\mu} \mathcal{A}^n_\nu - \partial_{\mathcal{R}_\nu} \mathcal{A}^n_\mu$.

Despite the striking resemblance between Equations (2.7) and (2.8), it is not trivial to formally show the correspondence between the Berry and the Gaussian curvatures. For the case of a torus ($g = 1$), a proof of their equivalence can be found in Yang et al. (2019).

3 Linear Response Theory

Linear response theory deals with the calculation of the response of a system to a small time-dependent perturbation acting on it when that response can be expanded in a power series of the strength of the perturbation, such that to first order, it is a linear function.

This chapter - mainly based on Sections 3 and 4 from Giuliani, Vignale (2005) - starts with the basis of linear response theory and builds its way to the emergence of electrical conductivity, as the current response to an applied vector potential.

3.1 General Theory

The Hamiltonian of a system which is acted upon by an external time-dependent field $F(t)$ linearly coupled to an observable $\hat{B}$ is given by $\hat{H}_F(t) = \hat{H} + F(t)\hat{B}$, where $\hat{H}$ is the time-independent Hamiltonian and $F(t)$ vanishes for t earlier than an initial time t_0.

Our goal is to calculate the expectation value of an observable $\hat{A}$ under the influence of the perturbation. Since the perturbation only starts acting at t_0, we can assume the system was in thermal equilibrium until then[1], such that the expectation value of $\hat{A}$ is given by $\langle \hat{A} \rangle_0 = \sum_n P_n \langle \psi_n | \hat{A} | \psi_n \rangle$, where P_n is the usual occupation probability $P_n = e^{-\beta \varepsilon_n}/\mathcal{Z}$.

For an adiabatic process[2] we can write the time evolution of a state $|\psi_n\rangle$, plug it into the Schrödinger Equation and then approximate it to first order in F:

$$|\psi_n(t)\rangle = \hat{U}(t, t_0) |\psi_n(t_0)\rangle = e^{-i\hat{H}(t-t_0)} \underbrace{\hat{U}_F(t, t_0)}_{\text{due to } F(t)} |\psi_n(t_0)\rangle , \tag{3.1a}$$

$$i\frac{\partial}{\partial t}\hat{U}_F(t, t_0) = F(t) \underbrace{\hat{B}(t - t_0)}_{\hat{B} = e^{i\hat{H}t}\hat{B}e^{-i\hat{H}t}} \hat{U}_F(t, t_0), \tag{3.1b}$$

$$\hat{U}_F(t, t_0) \approx e^{-i\hat{H}(t-t_0)} \left(\mathbb{1} - i \int_{t_0}^{t} \hat{B}(t' - t_0) F(t') dt' \right). \tag{3.1c}$$

The last equation is the cornerstone of linear response theory and allows us to evaluate the expectation value $\langle \hat{A} \rangle_{F(t)} = \sum_n P_n \langle \psi_n(t) | \hat{A} | \psi_n(t) \rangle$:

$$\langle \hat{A} \rangle_{F(t)} = \langle \hat{A} \rangle_0 - i \int_{t_0}^{t} \langle [\hat{A}(t), \hat{B}(t')] \rangle_0 F(t') \, dt'. \tag{3.2}$$

The last objective of this section is to introduce the retarded linear response function $\chi_{AB}(\tau)$, which describes the response of the observable $\hat{A}$ at time t to a perturbation that coupled to the observable $\hat{B}$ at an earlier time $t - \tau$ (Giuliani, Vignale, 2005)[3]:

$$\langle \hat{A} \rangle_1 (t) = \langle \hat{A} \rangle_{F(t)} - \langle \hat{A} \rangle_0 = \int_0^{\infty} \chi_{AB}(\tau) F(t - \tau) \, d\tau, \tag{3.3}$$

[1] This is a key assumption for our formalism to be valid.

[2] Note that we first defined the adiabatic process in Section 2.2, see Equation (2.2).

[3] The subscript "1" is used to emphasise that we are only considering linear contributions.

where we set the "switching on" t_0 of the perturbation to $-\infty$ and defined $\chi_{AB}(\tau)$:

$$\chi_{AB}(\tau) = -i\,\theta(\tau)\,\langle[\hat{A}(\tau), \hat{B}(0)]\rangle_0\,. \tag{3.4}$$

3.2 Periodic Perturbations

Since every well-behaved function of time can be written as a superposition of periodic functions via a Fourier transform, it is sufficient to only consider periodic perturbations from here on: $F(t) = F_\omega\, e^{-i\omega t} + c.c.$, where F_ω is a complex amplitude.

The difficulty of dealing with periodic perturbation comes from the fact that they are not vanishing for any t_0, which prevents us from assuming our system was in equilibrium at some earlier time. However, this can be adjusted by choosing a potential whose amplitude is slowly turned on: $F_\omega \to F_\omega\, e^{\eta t}$, where we need η to be positive for it to vanish at $t \to -\infty$.

Inserting this form of the perturbation into Equation (3.3) gives:

$$\langle\hat{A}\rangle_1(t) = \langle\hat{A}\rangle_1(\omega)e^{-i\omega t} + c.c. = \chi_{AB}(\omega)F_\omega e^{-i\omega t} + c.c.\,, \tag{3.5}$$

where we must take the limit $\eta \to 0^+$ in order to recover the response function:

$$\chi_{AB}(\omega) = -i\lim_{\eta\to 0}\int_0^\infty \langle[\hat{A}(\tau), \hat{B}(0)]\rangle_0\; e^{i(\omega+i\eta)\tau}d\tau. \tag{3.6}$$

In the eigenstates representation of the non-interacting Hamiltonian $|\psi_n\rangle$, the response function in the frequency domain takes the form:

$$\chi_{AB}(\omega) = \sum_{m,n}\frac{P_m - P_n}{\omega - \omega_{nm} + i\eta}A_{mn}B_{nm}, \tag{3.7}$$

where we introduced the notation $O_{nm} = \langle\psi_n|\,\hat{O}\,|\psi_m\rangle$ and $\omega_{nm} = \varepsilon_n - \varepsilon_m$.

This form of the response function is called the Lehmann representation and will be used throughout this thesis many times. A detailed proof is given in Appendix A.

3.3 Density-Density Response Function

Dealing with interactions, we are often interested in calculating the response of the number density operator $\hat{n}(\boldsymbol{r}) = \sum_i \delta(\boldsymbol{r} - \hat{\boldsymbol{r}}_{\boldsymbol{i}})$ at some point $\boldsymbol{r}$, to a scalar potential $V_{\text{ext}}(\boldsymbol{r'}, t)$ which couples linearly to the density $\hat{n}(\boldsymbol{r'})$ at some other point $\boldsymbol{r'}$. This is called the density-density response function and in real space it is denoted by $\chi_{nn}(\boldsymbol{r}, \boldsymbol{r'}, t) \equiv \chi_{n(\boldsymbol{r})n(\boldsymbol{r'})}(t)$.

Similar to Equation (3.5), the expectation value of the number density operator is:

$$\langle n\rangle_1(\boldsymbol{q}, \omega) = \sum_{\boldsymbol{q'}}\chi_{nn}(\boldsymbol{q}, \boldsymbol{q'}, \omega)V_{\text{ext}}(\boldsymbol{q'}, \omega), \tag{3.8}$$

where we dropped the $+\,c.c.$ for conciseness and we defined[4]:

$$\chi_{nn}(\boldsymbol{q}, \boldsymbol{q'}, \omega) = \frac{1}{L^{\text{D}}}\int d^{\text{D}}r\, e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}\int d^{\text{D}}r'\, e^{-i\boldsymbol{q'}\cdot\boldsymbol{r'}}\int_0^\infty dt\,\chi_{nn}(\boldsymbol{r}, \boldsymbol{r'}, t)\, e^{i\omega t}. \tag{3.9}$$

[4]The volume factor of L^{D} comes from the Fourier transform, where D is the number of dimensions.

The physical meaning of Equation (3.8) is that a perturbation with wave vector $\boldsymbol{q}'$ induces a modulation in the expectation value of the density for all the wave vectors $\boldsymbol{q}$ for which the response function $\chi_{nn}(\boldsymbol{q}, \boldsymbol{q}', \omega)$ is non-zero (Giuliani, Vignale, 2005).

For a translationally invariant system, the response function only depends on the difference $|\boldsymbol{r} - \boldsymbol{r}'|$, which makes the spatial integrals over $d^{\mathrm{D}}r$ and $d^{\mathrm{D}}r'$ in Equation (3.9) vanish for the cases when $\boldsymbol{q} \neq \boldsymbol{q}'$:

$$\chi_{nn}(\boldsymbol{q}, \boldsymbol{q}', \omega) = \chi_{nn}(\boldsymbol{q}, \omega)\delta_{\boldsymbol{q},\boldsymbol{q}'}. \tag{3.10}$$

Taking $\hat{A} = \hat{n}_{\boldsymbol{q}}$ and $\hat{B} = \hat{n}_{-\boldsymbol{q}}$[5], the Lehmann representation for the density-density response function can be written in the form:

$$\chi_{nn}(\boldsymbol{q}, \omega) = \frac{1}{L^{\mathrm{D}}} \sum_{m,n} \frac{P_m - P_n}{\omega - \omega_{nm} + i\eta} \left| (\hat{n}_{\boldsymbol{q}})_{mn} \right|^2, \tag{3.11}$$

where an important remark is that including the dependence on the wave vector in the response function introduces a Fourier normalisation factor: $\chi_{nn}(\boldsymbol{q}, \omega) = L^{-\mathrm{D}}\chi_{nn}(\omega)$.

For a homogeneous system of non-interacting electrons, it is more convenient to write the previous equation in its second quantisation form, using the creation and annihilation operators which diagonalise the non-interacting Hamiltonian $\hat{H} = \sum_\alpha \varepsilon_\alpha \hat{a}^\dagger_\alpha \hat{a}_\alpha$. Then, the single-particle operators are:

$$\hat{A} = \sum_{i=1}^{N} \hat{A}_i = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{a}^\dagger_\alpha \, \hat{a}_\beta \quad \text{and} \quad \hat{B} = \sum_{i=1}^{N} \hat{B}_i = \sum_{\gamma,\delta} B_{\gamma\delta} \, \hat{a}^\dagger_\gamma \, \hat{a}_\delta. \tag{3.12}$$

Following similar steps as for the proof of the Lehmann representation given in Appendix A, the non-interacting linear response function[6] takes the form:

$$\chi^{(0)}_{AB}(\omega) = \sum_{\alpha,\beta} \frac{n_\alpha - n_\beta}{\omega + \varepsilon_\alpha - \varepsilon_\beta + i\eta} A_{\alpha\beta} B_{\beta\alpha}, \tag{3.13}$$

where $n_\alpha = (e^{\beta(\varepsilon_\alpha - \mu)} + 1)^{-1}$ is the Fermi-Dirac occupation number of a state α.

3.4 Current-Current Response Function

To compute the electrical conductivity, we first need to calculate the current response function of a uniform electron system subject to a general vector potential $\boldsymbol{A}(\boldsymbol{r}, t)$. Note that with the right gauge transformation, any scalar potential $V(\boldsymbol{r}, t)$ can be replaced by a vector potential, which has the advantage that it couples linearly to the current $\boldsymbol{j}$. The perturbed Hamiltonian is given by:

$$\hat{H}_{\boldsymbol{A}}(t) = \frac{1}{2m} \sum_i \left[\hat{\boldsymbol{p}}_i + \frac{e}{c} \boldsymbol{A}\left(\hat{\boldsymbol{r}}_i, t\right) \right]^2 + U\left(\hat{\boldsymbol{r}}_1, \ldots \hat{\boldsymbol{r}}_N\right). \tag{3.14}$$

Considering only the linear terms in $\boldsymbol{A}(\boldsymbol{r}, t)$:

$$\hat{H}_{\boldsymbol{A}}(t) = \hat{H} + \frac{1}{2m} \sum_i \frac{e}{c} (\hat{\boldsymbol{p}}_i \cdot \boldsymbol{A} + \boldsymbol{A} \cdot \hat{\boldsymbol{p}}_i) = \hat{H} + \frac{e}{c} \int \hat{\boldsymbol{j}}_p(\boldsymbol{r}) \cdot \boldsymbol{A}(\boldsymbol{r}, t) d\boldsymbol{r}, \tag{3.15}$$

[5]Note that in Fourier space: $\hat{n}_{\boldsymbol{q}} = \sum_i e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i}$, so we have $\hat{A}^\dagger = \hat{B}$.

[6]We use the superscript "(0)" to emphasise the non-interactive aspect.

where the form of the Hamiltonian was simplified by introducing the paramagnetic component of the current density $\hat{\boldsymbol{j}}_p$ (Tao, 2005):

$$\hat{\boldsymbol{j}}_p(\boldsymbol{r}) = \frac{1}{2m}\sum_i \left(\hat{\boldsymbol{p}}_i \delta(\boldsymbol{r} - \hat{\boldsymbol{r}}_i) + \delta(\boldsymbol{r} - \hat{\boldsymbol{r}}_i)\hat{\boldsymbol{p}}_i \right), \tag{3.16}$$

$$\hat{\boldsymbol{j}}(\boldsymbol{r}) = \hat{\boldsymbol{j}}_p(\boldsymbol{r}) + \frac{e}{mc}\hat{n}(\boldsymbol{r})\boldsymbol{A}(\boldsymbol{r}). \tag{3.17}$$

The reason why $\hat{\boldsymbol{j}}_p$ is used instead of $\hat{\boldsymbol{j}}$ lies in the fact that the paramagnetic component of the current density does not depend on the vector potential $\boldsymbol{A}$, and therefore we can consider it an "intrinsic" quantity of the system, linearly coupled to the perturbation $\boldsymbol{A}(\boldsymbol{r}, t)$ (Giuliani, Vignale, 2005). Similar to Equation (3.8), we can write:

$$\langle \hat{j}_{p\alpha} \rangle_1 (\boldsymbol{q}, \omega) = \frac{e}{c}\sum_{\beta=1}^{3}\sum_{\boldsymbol{q}'} \chi_{j_{p\alpha} j_{p\beta}}(\boldsymbol{q}, \boldsymbol{q}', \omega) A_\beta(\boldsymbol{q}', \omega), \tag{3.18}$$

where α and β label Cartesian coordinates.

We are particularly interested in translationally invariant systems, for which the current-current response function obeys the same relation as the one given in Equation (3.10). Since the expectation value of the diamagnetic current - the second part of Equation (3.17) - is already linear in the perturbation, we can recover the physical current-current response function by adding this contribution to our previous equation:

$$\langle \hat{j}_\alpha \rangle (\boldsymbol{q}, \omega) = \frac{e}{c}\sum_{\beta=1}^{3} \chi^J_{\alpha\beta}(\boldsymbol{q}, \omega) A_\beta(\omega), \tag{3.19}$$

$$\chi^J_{\alpha\beta}(\boldsymbol{q}, \omega) = \frac{n}{m}\delta_{\alpha\beta} + \chi_{j_{p\alpha} j_{p\beta}}(\boldsymbol{q}, \omega). \tag{3.20}$$

3.4.1 The Kubo Formula

In order to calculate the response of an electronic system to a uniform electric field $\boldsymbol{E}(t)$, we introduce it in our formalism through $\boldsymbol{A}(t) = -c\int_0^t \boldsymbol{E}(t')dt'$, such that $\boldsymbol{A}(\omega) = -ic/\omega \boldsymbol{E}(\omega)$. For the case $q = 0$, if we plug this form of the perturbation into Equation (3.19), we get:

$$-ej_\alpha(0, \omega) = \frac{ie^2}{\omega}\sum_{\beta=1}^{3} \chi^J_{\alpha\beta}(0, \omega) E_\beta(\omega). \tag{3.21}$$

Since the conductivity σ is related to the electric field via $\boldsymbol{J} = -e\boldsymbol{j} = \sigma\boldsymbol{E}$, we determine the macroscopic electrical conductivity to be:

$$\sigma_{\alpha\beta}(\omega) = \frac{ie^2}{\omega}\chi^J_{\alpha\beta}(0, \omega) = \frac{ie^2}{\omega}\left[\frac{n}{m}\delta_{\alpha\beta} + \chi_{j_{p\alpha} j_{p\beta}}(0, \omega)\right], \tag{3.22}$$

which is known as the Kubo formula (Kubo, 1957).

To express the Hall conductivity from the transverse conductivity σ_{xy}, we must take the direct current limit in our perturbations $\omega \to 0$ (Murayama, 2008), such that:

$$\sigma_H = \lim_{\omega \to 0} \sigma_{xy}(\omega). \tag{3.23}$$

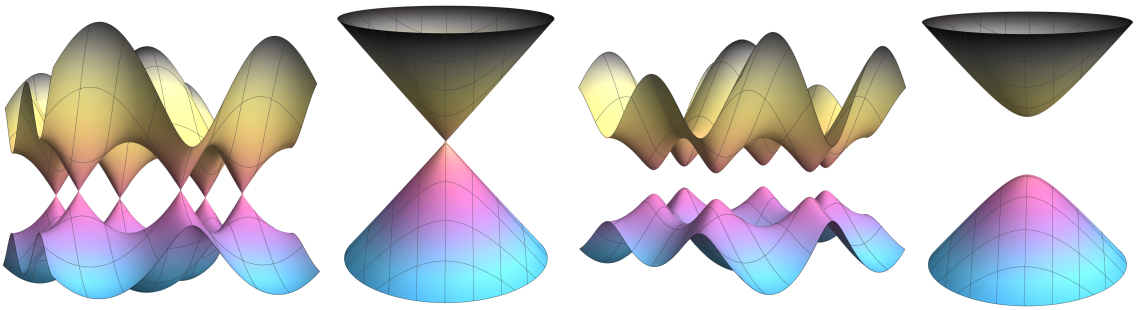
4 Massive Dirac Fermions

As a concrete example of the calculation of Hall conductivity by means of linear response theory, we are looking at massive Dirac fermions - particles with nonzero mass whose behaviour is described by the Dirac equation.

4.1 Connection to the Band Structure

Dirac fermions are known to emerge as effective low-energy quasiparticles in some band structures, graphene being the most famous example of a material hosting them (Neto et al., 2009). The energy spectrum close to the points K and K' in graphene (see Figure 2.1) consists of Dirac cones, which having a closed gap - Figure 4.1(a) - can be described by Dirac equation for $m = 0$. As a result, we refer to these particles as "massless".

A monolayer of hexagonal boron nitride has the same lattice structure as graphene (again, see Figure 2.1), but different on-site energies for the boron and the nitrogen atoms. This opens a gap 2Δ in the band structure - Figure 4.1(b) - which can be accounted for in the Dirac equation by adding a mass term proportional to the gap. In this case, we refer to the Dirac fermions as being "massive" (Fuchs, 2013).



(a) Band structure without a gap (b) Band structure with a gap

Figure 4.1: Tight binding band structure of a hexagonal lattice (first and third plot), also zoomed in around one of the K points (second and fourth plot).

In analogy to the Higgs mechanism where particles acquire mass by spontaneously breaking a symmetry, massive Dirac fermions can also be found in graphene whose inversion symmetry is broken. The underlying experimental mechanism for this requires a van der Waals heterojunction with a substrate that can produce an inhomogeneous electrostatic potential to break the symmetry of the sublattice at consecutive sites (Guan et al., 2017). Massive Dirac fermions can also be found in the Haldane model, where introducing a complex next-nearest-neighbour hopping gives rise to a "mass" term (Haldane, 1988).

4.2 Hamiltonian

The K-valley Hamiltonian of massive Dirac fermions confined to move in two dimensions is:

$$\hat{H} = v(\sigma_x \hat{p}_x + \sigma_y \hat{p}_y) + mv^2\sigma_z = v\,\boldsymbol{\sigma} \cdot \hat{\boldsymbol{p}} + \Delta\,\sigma_z, \tag{4.1}$$

where σ_x and σ_y are Pauli matrices, Δ is half the energy gap, and v is a constant which replaces the speed of light c in the traditional Dirac equation. Similar to the spinor structure of the wave function, massive Dirac fermions are characterised by an intrinsic "pseudospin", which labels the two energy bands: "up" ($\uparrow$) and "down" ($\downarrow$).

For brevity, we can write $\hat{H} = \boldsymbol{\sigma} \cdot \boldsymbol{a}$, where $\boldsymbol{a} = (v\,\hat{p}_x, v\,\hat{p}_y, \Delta)$ and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. The eigenvalues $\lambda_\pm$ and eigenvectors $|\lambda_\pm\rangle$ of the Hamiltonian are:

$$\lambda_\pm = \pm\sqrt{\Delta^2 + v^2p^2} = \pm\,\varepsilon_0, \tag{4.2a}$$

$$|\lambda_+\rangle = \begin{pmatrix} v(\hat{p}_x - i\hat{p}_y) \\ \sqrt{\Delta^2 + v^2p^2} - \Delta \end{pmatrix} \quad \text{and} \quad |\lambda_-\rangle = \begin{pmatrix} -v(\hat{p}_x - i\hat{p}_y) \\ \sqrt{\Delta^2 + v^2p^2} + \Delta \end{pmatrix}. \tag{4.2b}$$

Using $\boldsymbol{a} = \varepsilon_0\,(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, in spherical coordinates they become:

$$|\lambda_+\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)\,e^{i\phi} \end{pmatrix} \quad \text{and} \quad |\lambda_-\rangle = \begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2)\,e^{i\phi} \end{pmatrix}, \tag{4.3}$$

where it is important to note that the angles θ and ϕ are both functions of the momentum[1], such that $\theta = \arccos(\Delta/\varepsilon_0)$ and $\phi = \arccos(p_x/p) = \arcsin(p_y/p)$.

4.3 Current Operator

By introducing a vector potential to our Hamiltonian, minimal coupling gives $\boldsymbol{p} \to \boldsymbol{p} + e\boldsymbol{A}$, so $\hat{H} \to \hat{H} + e\,v\,\boldsymbol{\sigma} \cdot \boldsymbol{A}$. We already saw in Section 3.4 that the paramagnetic current operator $\hat{\boldsymbol{j}}_p$ couples linearly to $\boldsymbol{A}$, so we can easily identify it to be $\hat{\boldsymbol{j}}_p = v\,\boldsymbol{\sigma}$.

Since our final goal is to determine the transverse Hall conductivity σ_{xy}, we can see from the Kubo formula that the diamagnetic contribution vanishes because of δ_{xy}, so we are left only with the paramagnetic current-current response function. In order to determine the latter, we need to go back to Equation (3.13) and evaluate it for $\hat{A} = \hat{j}^{\,x}_{\boldsymbol{q}}$ and $\hat{B} = \hat{j}^{\,y}_{-\boldsymbol{q}}$. For brevity, we drop the subscript "p" from here on.

The second quantisation matrix elements of $j^{\,x}$ and $j^{\,y}$, in a basis of single-particle states characterised by wave vector and spin[2] $\alpha \to |\boldsymbol{k}', \lambda'\rangle$ and $\beta \to |\boldsymbol{k}, \lambda\rangle$ are:

$$j^{\,x}_{\alpha\beta} = \langle \boldsymbol{k}', \lambda' |\, j^{\,x}_{\boldsymbol{q}}\, | \boldsymbol{k}, \lambda \rangle = v\,\langle \psi^{\lambda'}_{\boldsymbol{k}'} |\, \sigma_x\, | \psi^{\lambda}_{\boldsymbol{k}} \rangle\, \delta_{\boldsymbol{k}', \boldsymbol{k}+\boldsymbol{q}}, \tag{4.4a}$$

$$j^{\,y}_{\beta\alpha} = \langle \boldsymbol{k}, \lambda |\, j^{\,y}_{-\boldsymbol{q}}\, | \boldsymbol{k}', \lambda' \rangle = v\,\langle \psi^{\lambda}_{\boldsymbol{k}} |\, \sigma_y\, | \psi^{\lambda'}_{\boldsymbol{k}'} \rangle\, \delta_{\boldsymbol{k}', \boldsymbol{k}+\boldsymbol{q}}, \tag{4.4b}$$

where we used $|\psi^{\lambda}_{\boldsymbol{k}}\rangle$ to denote the pseudospin eigenfunctions given in Equation (4.3), with λ taking values "+" or "-" to denote the upper or lower band, respectively. It is crucial to remember that they depend on their corresponding wave vectors, and therefore must pay attention to the fact that $\theta(\boldsymbol{k}) \neq \theta(\boldsymbol{k}')$ and $\phi(\boldsymbol{k}) \neq \phi(\boldsymbol{k}')$.

Putting everything into Equation (3.13), the transverse current response has the form[3]:

$$\chi_{j^x j^y}(q, \omega) = \frac{1}{L^{\mathrm{D}}}\, v^2 \sum_{\substack{\boldsymbol{k}, \boldsymbol{k}' \\ \lambda, \lambda'}} \frac{n^{\lambda'}_{\boldsymbol{k}'} - n^{\lambda}_{\boldsymbol{k}}}{\omega + \varepsilon^{\lambda'}_{\boldsymbol{k}'} - \varepsilon^{\lambda}_{\boldsymbol{k}} + i\eta}\; \langle \psi^{\lambda'}_{\boldsymbol{k}'} |\, \sigma_x\, | \psi^{\lambda}_{\boldsymbol{k}} \rangle\, \langle \psi^{\lambda}_{\boldsymbol{k}} |\, \sigma_y\, | \psi^{\lambda'}_{\boldsymbol{k}'} \rangle\, \delta_{\boldsymbol{k}', \boldsymbol{k}+\boldsymbol{q}}\,, \tag{4.5}$$

[1] We can use $\boldsymbol{p}$ and $\boldsymbol{k}$ interchangeably, as long as we introduce the factors of $\hbar$ in our end results.

[2] Here we switched the notation of the spin from σ to λ to avoid confusion with the Pauli matrices.

[3] Here we went from $\chi_{j^x j^y}(\omega)$ to $\chi_{j^x j^y}(q, \omega)$, while getting a Fourier normalisation factor of $L^{-\mathrm{D}}$.

where at zero temperature, the Fermi-Dirac occupation number reduces to a step function.

For the case $q = 0$, the algebra is greatly simplified as $\delta_{\boldsymbol{k}',\boldsymbol{k}}$ allows us to evaluate the elements $\langle\psi^{\lambda'}_{\boldsymbol{k}'}|\,\sigma_x\,|\psi^{\lambda}_{\boldsymbol{k}}\rangle\,\langle\psi^{\lambda}_{\boldsymbol{k}}|\,\sigma_y\,|\psi^{\lambda'}_{\boldsymbol{k}'}\rangle$ for all spin combinations, using the same angles:

$$\langle+|\,\sigma_x\,|+\rangle\,\langle+|\,\sigma_y\,|+\rangle = \sin^2\theta\cos\phi\sin\phi, \tag{4.6a}$$
$$\langle+|\,\sigma_x\,|-\rangle\,\langle-|\,\sigma_y\,|+\rangle = i\cos\theta - \sin^2\theta\cos\phi\sin\phi, \tag{4.6b}$$
$$\langle-|\,\sigma_x\,|+\rangle\,\langle+|\,\sigma_y\,|-\rangle = -i\cos\theta - \sin^2\theta\cos\phi\sin\phi, \tag{4.6c}$$
$$\langle-|\,\sigma_x\,|-\rangle\,\langle-|\,\sigma_y\,|-\rangle = \sin^2\theta\cos\phi\sin\phi. \tag{4.6d}$$

4.4 Calculation of Hall Conductivity

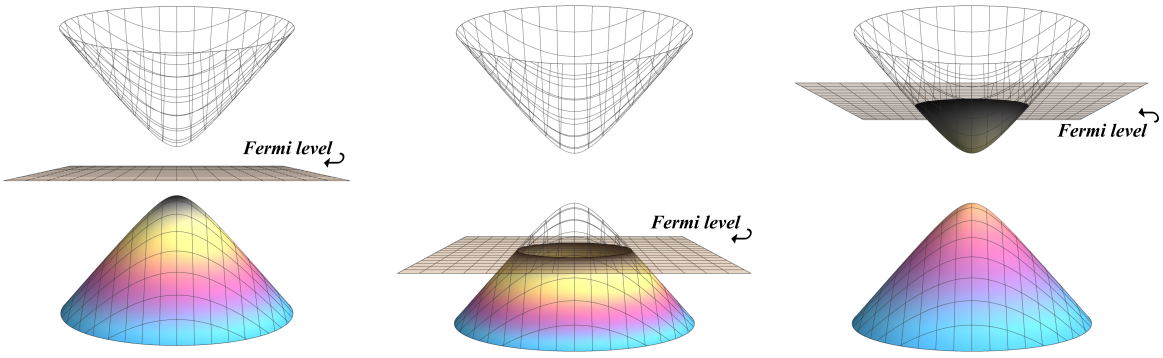
After performing the sum over the spins and using the expressions from Equations (4.6), we bring our equation for $\chi_{j^x j^y}(0,\omega)$ to a compact form:

$$\chi_{j^x j^y}(0,\omega) = \frac{v^2}{L^{\mathrm{D}}}\sum_{\boldsymbol{k}}(n^+_{\boldsymbol{k}} - n^-_{\boldsymbol{k}})\left(\frac{(\sin^2\theta\sin 2\phi)/2 + i\cos\theta}{\omega - 2\varepsilon_{\boldsymbol{k}} + i\eta} - \frac{(\sin^2\theta\sin 2\phi)/2 - i\cos\theta}{\omega + 2\varepsilon_{\boldsymbol{k}} + i\eta}\right). \tag{4.7}$$

We evaluate this response function for 3 different scenarios - Figure 4.2: when the Fermi level is inside the band structure gap (a), below (b) or above it (c). When a state defined by wave vector $\boldsymbol{k}$ and pseudospin λ is occupied (i.e. below the Fermi energy), we have $n^{\lambda}_{\boldsymbol{k}} = 1$, while for the case when it is empty (i.e. above the Fermi energy) $n^{\lambda}_{\boldsymbol{k}} = 0$.

In order to compute this, we replace the sum over the wave vectors with an integral:

$$\sum_{\boldsymbol{k}} f(\boldsymbol{k}) \xrightarrow{\text{in } 2D} \frac{L^{\mathrm{D}}}{(2\pi)^2}\int d^2k\, f(\boldsymbol{k}) = \frac{L^{\mathrm{D}}}{(2\pi)^2}\int_0^{\infty} dk\, k \int_0^{2\pi} d\phi\, f(k,\phi). \tag{4.8}$$



(a) ε_F inside the gap (b) ε_F below the gap (c) ε_F above the gap

Figure 4.2: The band structure around the Dirac point K with the Fermi level set inside, below or above the gap. The coloured regions represent the occupied states.

4.4.1 Fermi Energy Inside the Gap

In this case, we have $n^+_{\boldsymbol{k}} = 0$ and $n^-_{\boldsymbol{k}} = 1$, for all wave vectors $\boldsymbol{k}$. Making use of the expressions in Section 4.2 and finally putting back our factors of $\hbar$, we arrive at the result:

$$\chi_{j^x j^y}(0,\omega) = \frac{i\,\Delta}{4\pi\hbar^2}\ln\left|\frac{2\Delta + \hbar\omega + i\eta}{2\Delta - \hbar\omega - i\eta}\right|, \tag{4.9}$$

which after taking the limit $\eta \to 0$ and expanding the logarithm to first order in ω gives:

$$\sigma_H = \frac{ie^2}{\not{\omega}} \cdot \frac{i\not{\omega}}{4\pi\hbar} = -\frac{1}{2}\frac{e^2}{h}. \tag{4.10}$$

4.4.2 Fermi Energy Below the Gap

In this case, we have $n_{\boldsymbol{k}}^{+} = 0$ for all $\boldsymbol{k}$, but $n_{\boldsymbol{k}}^{-} = \theta(k - k_F)$, which changes the limit of integration from $(0, \infty)$ to (k_F, ∞). Taking the same limits as before, we get:

$$\sigma_H = -\frac{\Delta}{2\sqrt{v^2\,\hbar^2\,k_F^2 + \Delta^2}}\,\frac{e^2}{h}. \tag{4.11}$$

4.4.3 Fermi Energy Above the Gap

In this case we have $n_{\boldsymbol{k}}^{-} = 1$ for all $\boldsymbol{k}$, but $n_{\boldsymbol{k}}^{+} = \theta(k_F - k)$, which gives the full contribution from the lower band - Equation (4.10), plus the one from the upper band with the new limit of integration $(0, k_F)$. Adding the two and taking the same limits as before, we get an expression which is identical to the Hall conductivity from Equation (4.11).

These results imply that the contribution of the holes when the Fermi level is below the gap (b) is in fact identical to the contribution of the electrons when the Fermi level is (symmetrically) above the gap (c), both these cases characterising a conductor[4].

4.5 Berry Curvature and Conductivity

Since only the real part of the conductivity is of physical importance, Equation (3.22) allows us to write $\mathrm{Re}(\sigma_H) = \lim_{\omega\to 0}\left[-\frac{e^2}{\omega}\,\mathrm{Im}(\chi_{j^x j^y}(0,\omega))\right]$. The latter can be calculated from Equation (4.5), by separating it into two sums, relabelling $\lambda \leftrightarrow \lambda'$ in the second and then taking the common factor out:

$$\sigma_H = -\frac{2e^2}{\hbar}\frac{1}{L^{\mathrm{D}}}\sum_{\boldsymbol{k},\lambda,\lambda'}\frac{n_{\boldsymbol{k}}^{\lambda}}{(\varepsilon_{\boldsymbol{k}}^{\lambda} - \varepsilon_{\boldsymbol{k}}^{\lambda'})^2}\,\mathrm{Im}\left[\langle\psi_{\boldsymbol{k}}^{\lambda}|\,j^x\,|\psi_{\boldsymbol{k}}^{\lambda'}\rangle\,\langle\psi_{\boldsymbol{k}}^{\lambda'}|\,j^y\,|\psi_{\boldsymbol{k}}^{\lambda}\rangle\right]. \tag{4.12}$$

For our massive Dirac fermions Hamiltonian - Equation (4.1) - we can use the relation $\hat{\boldsymbol{j}} = v\boldsymbol{\sigma}$ and thus write $\hat{\boldsymbol{j}} = \nabla_{\boldsymbol{k}}\hat{H}$. After calculating the matrix elements in this form[5]:

$$\langle\psi_{\boldsymbol{k}}^{\lambda}|\,\nabla_{\boldsymbol{k}}\hat{H}\,|\psi_{\boldsymbol{k}}^{\lambda'}\rangle = \underbrace{\nabla_{\boldsymbol{k}}\,\langle\psi_{\boldsymbol{k}}^{\lambda}|\,\hat{H}\,|\psi_{\boldsymbol{k}}^{\lambda'}\rangle}_{=0} - \underbrace{\langle\nabla_{\boldsymbol{k}}\,\psi_{\boldsymbol{k}}^{\lambda}|\,\hat{H}\,|\psi_{\boldsymbol{k}}^{\lambda'}\rangle - \langle\psi_{\boldsymbol{k}}^{\lambda}|\,\hat{H}\,|\nabla\boldsymbol{k}\,\psi_{\boldsymbol{k}}^{\lambda'}\rangle}_{=(\varepsilon_{\boldsymbol{k}}^{\lambda'}-\varepsilon_{\boldsymbol{k}}^{\lambda})\langle\psi_{\boldsymbol{k}}^{\lambda}|\nabla_{\boldsymbol{k}}\psi_{\boldsymbol{k}}^{\lambda'}\rangle = (\varepsilon_{\boldsymbol{k}}^{\lambda}-\varepsilon_{\boldsymbol{k}}^{\lambda'})\langle\nabla_{\boldsymbol{k}}\psi_{\boldsymbol{k}}^{\lambda}|\psi_{\boldsymbol{k}}^{\lambda'}\rangle}, \tag{4.13}$$

we can perform the sum over λ' in Equation (4.12) and therefore arrive at the relation:

$$\sigma_H = -\frac{2e^2}{\hbar}\frac{1}{L^{\mathrm{D}}}\sum_{\boldsymbol{k},\lambda} n_{\boldsymbol{k}}^{\lambda}\underbrace{\mathrm{Im}\left[\langle\nabla_{\boldsymbol{k}}^{x}\psi_{\boldsymbol{k}}^{\lambda}|\nabla_{\boldsymbol{k}}^{y}\psi_{\boldsymbol{k}}^{\lambda}\rangle\right]}_{=-\Omega^n/2}. \tag{4.14}$$

For a fully occupied band - Figure 4.2(a) - we have $n_{\boldsymbol{k}}^{\lambda} = 1$, which leaves us with the integral over the Berry curvature, after replacing the sum with an integral, as shown in

[4] Our results from all three cases are consistent with the calculations performed in Sinitsyn et al. (2006).

[5] We use the orthonormality of the eigenvectors $\nabla_{\boldsymbol{k}}\left(\langle\psi_{\boldsymbol{k}}^{\lambda}|\psi_{\boldsymbol{k}}^{\lambda'}\rangle\right) = 0$.

Equation (4.8). Using Chern's theorem we get $\sigma_H = C_1 e^2/h$, which explicitly shows that the Hall conductivity is quantised (Niu et al., 1985).

However, comparing this with our result from Subsection 4.4.1, it seems surprising that the Hall conductivity is only half-integer. This is due to the fact that we only considered the K-valley Hamiltonian throughout our calculations, and it is the sum of the K and K' valley conductivities the physical one that is quantised (Sinitsyn et al., 2006).

For a system that breaks time-reversal symmetry, like the Haldane model (Haldane, 1988), we can include the contribution from the K'-valley by performing the same steps we did in this chapter, but for the Hamiltonian $\hat{H} = v(-\sigma_x \hat{p}_x + \sigma_y \hat{p}_y) - mv^2\sigma_z$ (Girvin, Yang, 2019), noting that the current operators in this case are $\hat{j}^x \to -v\sigma_x$ and $\hat{j}^y \to v\sigma_y$. Since this yields exactly the same matrix elements as the ones given in the four expressions from (4.6), we can conclude that the Hall conductivity is equal in the K and K' valleys[6]. Taking into account both these contributions, we get our final result $\sigma_H = -e^2/h$, which proves that the Hall conductivity is indeed quantised, with Chern number $C_1 = -1$.

This outcome hints to an important remark: the Hall conductivity σ_H has both a topological (quantised) part, due to the filled band, and a non-topological part, due to the doping of the material. Since this is only valid when the Fermi level is inside the gap, we can now clarify our interest in massive rather than massless Dirac fermions.

5 Perturbative Linear Response Functions

The goal of this project is to analyse the impact of interactions on both the topological and non-topological part of the Hall conductivity, provided these interactions act as "continuous deformations" and therefore preserve the topology of the system (i.e. do not close the gap). By treating these interactions as perturbations, we want to calculate their first-order contribution on the conductivity by using a diagrammatic approach.

5.1 Overview of Perturbative Calculation

Starting with a time-independent Hamiltonian, we can separate its interaction such that $\hat{H} = \hat{H}_0 + \hat{H}_I$, where $\hat{H}_0$ is the Hamiltonian of the non-interacting system and the interaction is solely enclosed in $\hat{H}_I$. The final goal of this chapter is to expand the response function - Equation (3.4) - into a power series of the coupling constant of the interaction.

This formulation of perturbation theory is greatly simplified in the Dirac (interaction) picture, where using a time-ordered correlation function proves to be more convenient than our previous approach (Giuliani, Vignale, 2005):

$$\chi^T_{AB}(t) = -i \left\langle \mathcal{T}[\hat{A}(t)\hat{B}(0)] \right\rangle_0 , \tag{5.1}$$

where we can always go back to our response function from Chapter 3 using:

$$\mathrm{Re}[\chi_{A^\dagger A}(\omega)] = \mathrm{Re}[\chi^T_{A^\dagger A}(\omega)], \tag{5.2a}$$

$$\mathrm{Im}[\chi_{A^\dagger A}(\omega)] = \mathrm{sign}(\omega)\, \mathrm{Im}[\chi^T_{A^\dagger A}(\omega)]. \tag{5.2b}$$

The proof of this, together with a graphical representation of the correspondence between the response and the time-ordered correlation functions, are given in Appendix C.

[6] This is not valid for a system with time-reversal symmetry, like graphene with a gap, as the mass term has the same sign in both K and K' valleys. In this case, the Berry curvature and the Hall conductivity are "odd" vis-a-vis the two valleys, so the total contribution to σ_H vanishes (Xiao et al., 2010).

5.2 Adiabatic Assumption

The central assumption for perturbation theory is that the interaction is slowly and continuously being "switched on", such that at $t \to \pm\infty$, the interacting and non-interacting ground states coincide. A good candidate for a Hamiltonian describing this perturbation is:

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_I\, e^{-\eta|t|}, \tag{5.3}$$

where it is important to note that within a "time window" around $t = 0$, $\hat{H}(t)$ is identical to our physical, observable Hamiltonian (Giuliani, Vignale, 2005).

In the Dirac picture, the time evolution of the states is entirely due to interactions, while operators propagate in time with the non-interacting Hamiltonian. In our correlation function, we are taking the expectation value of the operators in the ground state $|\Phi_0\rangle$, which was established to coincide with the ground state of the interacting Hamiltonian at $t_0 \to -\infty$. In order to evaluate it in the Dirac picture, we need to "propagate" the Heisenberg operators $\hat{A}$ and $\hat{B}$ from t_0 to t and $t = 0$, respectively[1]:

$$\begin{aligned}\hat{A}(t)\hat{B}(0) &= \hat{U}_I^{-1}(t,-\infty)\hat{A}_I(t)\hat{U}_I(t,-\infty)\hat{U}_I^{-1}(0,-\infty)\hat{B}_I(0)\hat{U}_I(0,-\infty) \\ &= \hat{U}_I(-\infty,t)\,\hat{A}_I(t)\,\hat{U}_I(t,-\infty)\,\hat{U}_I(-\infty,0)\,\hat{B}_I(0)\,\hat{U}_I(0,-\infty).\end{aligned} \tag{5.4}$$

Time-ordering our operators in the correlation function and making use of the property $\hat{U}_I(t_3,t_1) = \hat{U}_I(t_3,t_2)\,\hat{U}_I(t_2,t_1)$, we can write[2]:

$$\chi_{AB}^T(t) = -i\,\langle\Phi_0|\,\hat{U}_I(-\infty,\infty)\mathcal{T}[\hat{A}_I(t)\hat{B}\,\hat{U}_I(\infty,-\infty)]\,|\Phi_0\rangle\,. \tag{5.5}$$

Looking back at the way we defined our adiabatic evolution, the ground state $|\Phi_0\rangle$ propagating from $t \to -\infty$ to $t \to \infty$ can only acquire a phase: $\hat{U}_I(\infty,-\infty)\,|\Phi_0\rangle = e^{i\alpha}\,|\Phi_0\rangle$, which is in fact the expectation value of the scattering matrix $\hat{S}$:

$$e^{i\alpha} = \langle\Phi_0|\,\hat{U}_I(\infty,-\infty)\,|\Phi_0\rangle = \langle\Phi_0|\,\hat{S}\,|\Phi_0\rangle\,, \tag{5.6a}$$

$$\hat{S} = \hat{U}_I(\infty,-\infty) = \mathcal{T}\exp\Big(-i\int_{-\infty}^{\infty}\hat{H}_I(t')e^{-\eta|t'|}dt'\Big). \tag{5.6b}$$

Plugging this into Equation (5.5), we get our expression for the correlation function:

$$\chi_{AB}^T(t) = -i\,\frac{\langle\Phi_0|\,\mathcal{T}[\hat{A}_I(t)\,\hat{B}\,\hat{S}]\,|\Phi_0\rangle}{\langle\Phi_0|\,\hat{S}\,|\Phi_0\rangle}\,. \tag{5.7}$$

5.3 Non-interacting Green's Functions

In this formalism, calculating the correlation function comes down to evaluating the ground state averages that appear in the numerator. By the end of this section, our choice of working with a time-ordered correlation instead of the response function will be justified.

A contraction of two operators is defined as the difference between their time and normal ordering: $\hat{A}\hat{B} = \mathcal{T}[\hat{A}\hat{B}] - \mathcal{N}[\hat{A}\hat{B}]$. Since any field operator $\hat{\phi}$ contains an annihilation

[1] $\hat{U}_I$ gives the time evolution due to interactions alone: $\hat{U}(t,-\infty) = e^{-i\hat{H}_0 t}\hat{U}_I(t,-\infty)$.

[2] We dropped the subscript "I" for the operator $\hat{B}$ as $\hat{B}_I(t=0) = e^{i\hat{H}_0\cdot 0}\hat{B}\,e^{-i\hat{H}_0\cdot 0} = \hat{B}$.

$\hat{\phi}^-$ and a creation $\hat{\phi}^+$ part, for $t > 0$, $\hat{A}(t)\hat{B}(0) \equiv \hat{A}\hat{B}$, we have[3]:

$$\mathcal{T}[\hat{A}\hat{B}] = (\hat{A}^+ + \hat{A}^-)(\hat{B}^+ + \hat{B}^-) = \hat{A}^+\hat{B}^+ + \hat{A}^-\hat{B}^- + \hat{A}^-\hat{B}^+ + \hat{A}^+\hat{B}^-, \tag{5.8a}$$

$$\mathcal{N}[\hat{A}\hat{B}] = \hat{A}^+\hat{B}^+ + \hat{A}^-\hat{B}^- + \hat{B}^+\hat{A}^- + \hat{A}^+\hat{B}^-, \tag{5.8b}$$

so the contraction is just a complex number $\overline{\hat{A}\hat{B}} = [\hat{A}^-, \hat{B}^+]$ (Lancaster, Blundell, 2014).

Wick's theorem (Wick, 1950) states that the time-ordering of a string of operators is equal to the normal ordering of the string itself, together with that of all the possible contractions of the operators: $\mathcal{T}[\hat{A}\hat{B}\ldots\hat{Z}] = \mathcal{N}\left[\hat{A}\hat{B}\ldots\hat{Z} + \text{all contractions of } \hat{A}\hat{B}\ldots\hat{Z}\right]$. When taking the expectation value of this in the ground state, it becomes clear that the only remaining terms are those whose operators are all paired in contractions.

Therefore, in order to calculate the (ground state) expectation value of a time-ordered string of operators, one must take the sum[4] of the products of the expectation values of the time-ordered possible pairs[5]. These non-vanishing contractions are known as non-interacting Green's functions or Feynman propagators:

$$G^{(0)}_{\alpha\beta}(t_1 - t_2) = -i\,\langle\Phi_0|\,T[\,\hat{a}_{\alpha I}(t_1)\,\hat{a}^\dagger_{\beta I}(t_2)]\,|\Phi_0\rangle\,, \tag{5.9}$$

and give the probability amplitude for an electron added to the non-interacting Fermi sea in state β at time t_2 to be found in the state α at some later time t_1 - for $t_1 > t_2$. If $t_2 > t_1$, $G^{(0)}_{\alpha\beta}(t_1 - t_2)$ gives the corresponding probability amplitude for a hole created within the Fermi sea in state α at t_1 to be found in state β at t_2 (Giuliani, Vignale, 2005).

Similar to our approach in Chapter 3, it proves to be more convenient to work in Fourier space, where the non-interacting Green's function is:

$$G^{(0)}_{\alpha\beta}(\omega) = \int_{-\infty}^{\infty} G^{(0)}_{\alpha\beta}(t)\,e^{i\omega t}e^{-\eta|t|}dt = \delta_{\alpha\beta}\left[\frac{1-n_\alpha}{\omega - \varepsilon_\alpha + i\eta} + \frac{n_\alpha}{\omega - \varepsilon_\alpha - i\eta}\right]. \tag{5.10}$$

This can be further simplified by noting that the end results require the limit $\eta \to 0$, meaning that the sign preceding $i\eta$ only says whether the function's poles are above or below the real axis. We can absorb this sign by defining $\eta_\alpha = \text{sign}(\varepsilon_\alpha - \varepsilon_F)\eta$, such that:

$$G^{(0)}_{\alpha\beta}(\omega) = \frac{\delta_{\alpha\beta}}{\omega - \varepsilon_\alpha + i\eta_\alpha}. \tag{5.11}$$

5.4 Diagrammatic Perturbation Theory

The motivation behind using a diagrammatic perturbative approach comes from the fact that it provides a very suggestive graphical representation of our physical process, in terms of free electron propagators and electron-electron interactions (Giuliani, Vignale, 2005).

Returning to the initial problem of evaluating the terms in the correlation function - Equation (5.7) - we can expand the S-matrix from Equation (5.6b) in power series:

$$\langle\Phi_0|\mathcal{T}[\hat{A}_I(t)\hat{B}\hat{S}]|\Phi_0\rangle = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int dt_1 \ldots \int dt_n \langle\Phi_0|\mathcal{T}[\hat{A}_I(t)\hat{B}\hat{H}_{I\eta}(t_1)\ldots\hat{H}_{I\eta}(t_n)]|\Phi_0\rangle\,, \tag{5.12}$$

$$\langle\Phi_0|\hat{S}|\Phi_0\rangle = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int dt_1 \ldots \int dt_n \langle\Phi_0|\mathcal{T}[\hat{H}_{I\eta}(t_1)\ldots\hat{H}_{I\eta}(t_n)]|\Phi_0\rangle\,, \tag{5.13}$$

[3]Choosing $t < 0$ gives a similar result: $\overline{\hat{A}\hat{B}} = [\hat{B}^-, \hat{A}^+]$ (Peskin, 2018).

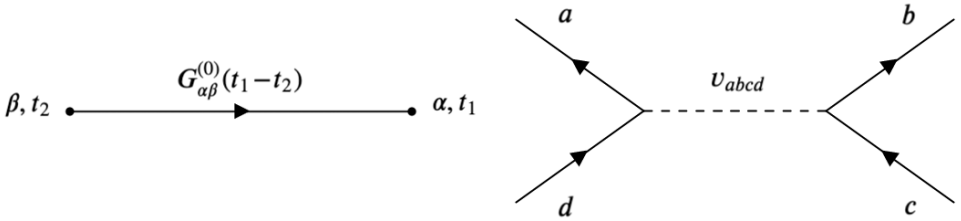
[4]For fermionic operators, the pairs requiring an odd number of permutations get an extra minus sign.

[5]Note that the expectation value of a contraction is equal to the expectation value of its time-ordering.

where we used the shorthand notation $\hat{H}_{I\eta}(t) = \hat{H}_I(t)\, e^{-\eta|t|}$. At this stage, we are only interested in the contribution of the interactions up to first order, so $n=0$ and $n=1$.

For the simple case of two-body interaction, the second quantisation form of the interacting Hamiltonian can be written as $\hat{H}_I = \frac{1}{2}\sum_{\text{abcd}} v_{abcd}\, \hat{a}_a^\dagger\, \hat{a}_b^\dagger\, \hat{a}_c\, \hat{a}_d$, where these operators describing the interaction are referred to as "internal". The operators $\hat{A}$ and $\hat{B}$ are the ones given in Equation (3.12) and are called "external"[6].

After plugging the second quantised operators back into Equations (5.12) and (5.13), and using our result from Wick's theorem, we are left with the sum of the products of all the Feynman propagators that can be obtained by pairing a string of 4 external and $4n$ internal operators. Graphically, this is equivalent to combining n interaction lines - Figure 5.1(b) - with $2n+2$ oriented solid lines - Figure 5.1(a) - in all possible ways, under the condition that solid lines cannot reverse their direction at any point. These are called Feynman diagrams (Peskin, 2018).



(a) Non-interacting Green's function (b) Coulomb interaction

Figure 5.1: Graphical representation of a Feynman propagator from state β to α (a) and a bare two-body interaction (b), where v_{abcd} is a Coulomb interaction matrix element.

5.5 Feynman Diagrams

As we already saw in the previous chapters, the physical properties of most systems are usually easier to understand in the Fourier space. In our case as well, the use of the Fourier representation for the non-interacting Green's functions $G^{(0)}_{\alpha\beta}(t) = \int d\omega/2\pi\, G^{(0)}_{\alpha\beta}(\omega) e^{-i\omega t}$ helps us gain insight into the intuition behind Feynman diagrams.

Since in the time domain at each vertex there is a corresponding time t_i which is later integrated over - Equation (5.12) - once written in Fourier, it gives rise to a delta function containing as many positive frequencies as solid lines exiting the vertex and as many negative ones as solid lines entering in that vertex. This implies that at each internal vertex, the sum of the incoming frequencies equals the sum of the outgoing frequencies. To make this frequency conservation valid at every vertex, either internal or external, we can assign a frequency ω at $\hat{A}$ and $-\omega$ at $\hat{B}$.

Similarly, for a translationally invariant system where the eigenstates of the non-interacting Hamiltonian are plane waves labelled $|\boldsymbol{k}, \lambda\rangle$, the non-interacting Green's function takes the form $G^{(0)}_{\boldsymbol{k},\lambda}(\omega) \equiv G^{(0)}_{\lambda}(\boldsymbol{k}, \omega)$[7]. The main advantage of choosing this wave vector representation is that complementary to working in the frequency domain, it also conserves momentum at each vertex, provided we assign a wave vector $\boldsymbol{q}$ at $\hat{A}$ and $-\boldsymbol{q}$ at $\hat{B}$.

For the first-order diagrams, the Feynman rules for a translationally invariant system can be found in Appendix B - based on Section 6.2.6 from Giuliani, Vignale (2005).

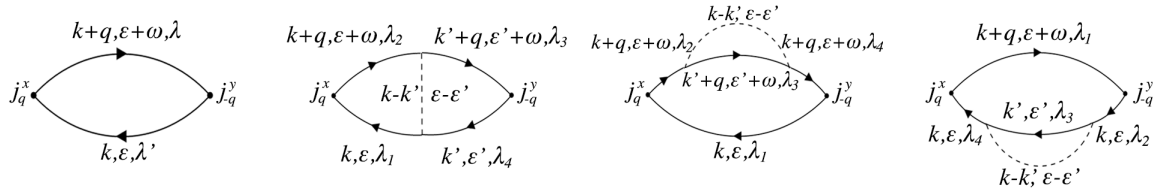
[6]Note that we are using Greek letters to label external operators, and Latin ones for internal operators.

[7]Since ω is in fact the energy, in this step we only switched to a dependence on the full 4-momentum.

5.5.1 Connected, Topologically Distinct Diagrams

Out of the $(2n+2)!$ ways of pairing creation and annihilation operators, many of the diagrams are completely negligible, as they do not contribute to the response function. The aim of this subsection is to identify the diagrams relevant to the cases $n = 0$, $n = 1$.

A diagram whose creation and annihilation operators are fully contracted among themselves within a subset of $\hat{H}_I$ is called "disconnected", as this means that the interaction is isolated from the external vertices. The total contribution of these diagrams is proved to sum up to $\langle\Phi_0|\hat{S}|\Phi_0\rangle$ - see Section 6.2.4 from Giuliani, Vignale (2005) - which allows us to ignore the denominator in Equation (5.7), provided we only consider connected contraction schemes from here on: $\chi^T_{AB}(t) = -i\,\langle\Phi_0|\,\mathcal{T}[\hat{A}_I(t)\,\hat{B}\,\hat{S}]\,|\Phi_0\rangle$.



(a) Zero-th order diagram (b) Exchange energy (c) Self-energy 1 (d) Self-energy 2

Figure 5.2: The zero and first order Feynman diagrams.

We can see from Figure 5.1(b) that simultaneously interchanging $a \leftrightarrow b$ and $c \leftrightarrow d$ leaves any diagram unchanged. This can be accounted for by only considering "topologically distinct" diagrams, which are the ones not related to each other by a trivial permutation of state labels. At order n, there are $n!\,2^n$ topologically identical diagrams[8].

Among the topologically distinct, connected diagrams, some no not connect the operators $\hat{A}$ and $\hat{B}$ by propagators and are therefore called "disjoint". Since they do not link the two operators to each other, they cannot contribute to their correlation function.

For the zero-th order, there is only one topologically distinct diagram. Looking at the first order - there are 24 ways of pairing the creation operators ($\hat{a}^\dagger_a$, $\hat{a}^\dagger_b$, $\hat{a}^\dagger_\alpha$, $\hat{a}^\dagger_\gamma$) with the annihilation ones ($\hat{a}_c$, $\hat{a}_d$, $\hat{a}_\beta$, $\hat{a}_\delta$), out of which 4 pairings give rise to disconnected diagrams. From the remaining 20 connected diagrams, only 10 are topologically distinct and 4 of those are disjoint. Despite thinking that the remaining 6 all contribute to the correlation function, 2 of them prove to vanish because their interaction carries zero momentum and 1 does not have any contributions to the physical response function. The proof of this can be found in Sections 6.2.8 and 6.3.1 from Giuliani, Vignale (2005) for $\chi_{nn}(\boldsymbol{q},\omega)$[9]. For completeness, these 3 diagrams that we neglect can be seen in Appendix B, Figure B.1. Finally, the zero and first order diagrams that we are interested in for the Hartree-Fock correction are shown in Figure 5.2.

5.6 Zero and First Order Contributions

Using similar Feynman rules as the ones given in Appendix B, but disregarding any contributions from the interaction, we obtain the following form for the zero-th order

[8]We can disregard this factor by simultaneously ignoring the 1/2 from the interacting Hamiltonian and the $1/n!$ from the power series expansion of the S-matrix.

[9]We still need to properly examine the latter for the case of a transverse current response.

current-current correlation function[10]:

$$\chi^T = -iv^2 \sum_{\lambda,\lambda'} \int \frac{d\boldsymbol{k}}{(2\pi)^{\mathrm{D}}} \int \frac{d\varepsilon}{2\pi} G_{\lambda}^{(0)}(\boldsymbol{k}+\boldsymbol{q},\varepsilon+\omega) G_{\lambda'}^{(0)}(\boldsymbol{k},\varepsilon)\, \langle \psi_{\boldsymbol{k}'}^{\lambda'} | \sigma_x | \psi_{\boldsymbol{k}}^{\lambda} \rangle \langle \psi_{\boldsymbol{k}}^{\lambda} | \sigma_y | \psi_{\boldsymbol{k}'}^{\lambda'} \rangle \,. \tag{5.14}$$

After adjusting the signs of $i\eta_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda}$ and $i\eta_{\boldsymbol{k}}^{\lambda'}$ to ensure the poles lie in the upper half-plane[11], we perform the energy integral by contour integration and arrive at the same expression as the one given in Equation (4.5).

Since we are left only with energy terms of the exchange and self type, considering their contribution suggests a "Hartree-Fock"-like correction to the correlation function. Applying the Feynman rules in Appendix B on the three diagrams from Figure 5.2, we obtain the following correlation functions:

$$\begin{aligned}\chi^{\mathrm{EX}} = v^2 \sum_{\substack{\lambda_1,\lambda_2 \\ \lambda_3,\lambda_4}} \int \frac{d\boldsymbol{k}}{(2\pi)^{\mathrm{D}}} \int \frac{d\boldsymbol{k}'}{(2\pi)^{\mathrm{D}}} \int \frac{d\varepsilon}{2\pi} \int \frac{d\varepsilon'}{2\pi} v_{\boldsymbol{k}-\boldsymbol{k}'} G_{\lambda_1}^{(0)}(\boldsymbol{k},\varepsilon) G_{\lambda_2}^{(0)}(\boldsymbol{k}+\boldsymbol{q},\varepsilon+\omega) \times \\ \times G_{\lambda_3}^{(0)}(\boldsymbol{k}'+\boldsymbol{q},\varepsilon'+\omega) G_{\lambda_4}^{(0)}(\boldsymbol{k}',\varepsilon') \langle \psi_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda_2} | \sigma_x | \psi_{\boldsymbol{k}}^{\lambda_1} \rangle \langle \psi_{\boldsymbol{k}'}^{\lambda_4} | \sigma_y | \psi_{\boldsymbol{k}'+\boldsymbol{q}}^{\lambda_3} \rangle \langle \psi_{\boldsymbol{k}'+\boldsymbol{q}}^{\lambda_3} | \psi_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda_2} \rangle \langle \psi_{\boldsymbol{k}}^{\lambda_1} | \psi_{\boldsymbol{k}'}^{\lambda_4} \rangle .\end{aligned}$$

$$\begin{aligned}\chi^{\mathrm{SE1}} = v^2 \sum_{\substack{\lambda_1,\lambda_2 \\ \lambda_3,\lambda_4}} \int \frac{d\boldsymbol{k}}{(2\pi)^{\mathrm{D}}} \int \frac{d\boldsymbol{k}'}{(2\pi)^{\mathrm{D}}} \int \frac{d\varepsilon}{2\pi} \int \frac{d\varepsilon'}{2\pi} v_{\boldsymbol{k}-\boldsymbol{k}'} G_{\lambda_1}^{(0)}(\boldsymbol{k},\varepsilon) G_{\lambda_2}^{(0)}(\boldsymbol{k}+\boldsymbol{q},\varepsilon+\omega) \times \\ \times G_{\lambda_3}^{(0)}(\boldsymbol{k}'+\boldsymbol{q},\varepsilon'+\omega) G_{\lambda_4}^{(0)}(\boldsymbol{k}+\boldsymbol{q},\varepsilon'+\omega) \langle \psi_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda_2} | \sigma_x | \psi_{\boldsymbol{k}}^{\lambda_1} \rangle \langle \psi_{\boldsymbol{k}}^{\lambda_1} | \sigma_y | \psi_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda_4} \rangle \langle \psi_{\boldsymbol{k}'+\boldsymbol{q}}^{\lambda_3} | \psi_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda_2} \rangle \langle \psi_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda_4} | \psi_{\boldsymbol{k}'+\boldsymbol{q}}^{\lambda_3} \rangle .\end{aligned}$$

$$\begin{aligned}\chi^{\mathrm{SE2}} = v^2 \sum_{\substack{\lambda_1,\lambda_2 \\ \lambda_3,\lambda_4}} \int \frac{d\boldsymbol{k}}{(2\pi)^{\mathrm{D}}} \int \frac{d\boldsymbol{k}'}{(2\pi)^{\mathrm{D}}} \int \frac{d\varepsilon}{2\pi} \int \frac{d\varepsilon'}{2\pi} v_{\boldsymbol{k}-\boldsymbol{k}'} G_{\lambda_1}^{(0)}(\boldsymbol{k}+\boldsymbol{q},\varepsilon+\omega) G_{\lambda_2}^{(0)}(\boldsymbol{k},\varepsilon) \times \\ \times G_{\lambda_3}^{(0)}(\boldsymbol{k}',\varepsilon') G_{\lambda_4}^{(0)}(\boldsymbol{k},\varepsilon) \langle \psi_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda_1} | \sigma_x | \psi_{\boldsymbol{k}}^{\lambda_4} \rangle \langle \psi_{\boldsymbol{k}}^{\lambda_2} | \sigma_y | \psi_{\boldsymbol{k}+\boldsymbol{q}}^{\lambda_1} \rangle \langle \psi_{\boldsymbol{k}'}^{\lambda_3} | \psi_{\boldsymbol{k}}^{\lambda_2} \rangle \langle \psi_{\boldsymbol{k}}^{\lambda_4} | \psi_{\boldsymbol{k}'}^{\lambda_3} \rangle .\end{aligned}$$

For the exchange energy, the two frequency integrals are performed using contour integration: $\int d\varepsilon/2\pi G_{\lambda_1}^{(0)}(\boldsymbol{k},\varepsilon) G_{\lambda_2}^{(0)}(\boldsymbol{k}+\boldsymbol{q},\varepsilon+\omega)$ and $\int d\varepsilon'/2\pi G_{\lambda_3}^{(0)}(\boldsymbol{k}'+\boldsymbol{q},\varepsilon'+\omega) G_{\lambda_4}^{(0)}(\boldsymbol{k}',\varepsilon')$, which give similar results to our frequency integral for zero order. We can retrieve the response function by fixing the poles of the integrand - (5.2b) - and perform the wave vector integrals in Mathematica, using asymptotic integration to first order in ω.

Including any type of Coulomb interaction $v_{\boldsymbol{k}-\boldsymbol{k}'}$ gives rise to infrared divergences for $\boldsymbol{q}=0$, which we are unable to amend at this point. Possible solutions for removing these divergences in the future are the Pauli-Villars regularisation method, which introduces a fictions massive photon, and dimensional regularisation, where the integral over $\boldsymbol{k}$ is evaluated in a "$D+\xi$"-dimensional space, where $\xi \to 0$ (Peskin, 2018).

However, if we consider a contact interaction $v_{\boldsymbol{k}-\boldsymbol{k}'} = V$, valid for systems where only short-range interactions are relevant, similar to the Hubbard model (Hubbard, 1963), we obtain a non-divergent response function, when the Fermi level is inside the gap $\varepsilon_F = 0$:

$$\chi^{\mathrm{EX}} = -\frac{iV\pi^2\lambda^2 \left(-\Delta + \sqrt{\Delta^2+\lambda^2}\right)\omega}{\Delta^2+\lambda^2}, \tag{5.15}$$

where λ is the cutoff in both k and k', used to avoid the ultraviolet divergences.

[10] We omitted the subscript $j_{\boldsymbol{q}}^x j_{-\boldsymbol{q}}^y$ for brevity.

[11] We close the integral in the upper half-plane because of Step 9 in the Feynman rules.

For the self-energy diagrams, the integral over ε' is straightforward, as it contains only one Green's function: $\int d\varepsilon'/2\pi G^{(0)}_{\lambda_3}(\boldsymbol{k'}+\boldsymbol{q},\varepsilon'+\omega) = in^{\lambda_3}_{\boldsymbol{k'}+\boldsymbol{q}}$. However, the one over ε proves to be much harder to solve, since it requires two cases to be analysed separately: $\lambda_2 = \lambda_4$, which produces one double and one single pole, and $\lambda_2 \neq \lambda_4$, which gives three single poles. For contact interaction, the response function calculated in Mathematica is:

$$\chi^{\mathrm{SE}} = -\frac{iV\pi^2\Delta\lambda^2\left(-\Delta+\sqrt{\Delta^2+\lambda^2}\right)\omega}{(\Delta^2+\lambda^2)^{3/2}}, \tag{5.16}$$

where both self-energy diagrams were considered and their contributions added together.

The exchange and self-energy contributions add up to:

$$\chi = \frac{-iV\pi^2\lambda^4\omega}{(\Delta^2+\lambda^2)^{3/2}}, \tag{5.17}$$

which is clearly divergent as $\lambda \to \infty$. A possible solution for removing this divergence is to renormalise the mass term Δ or the velocity constant v (Tsvelik, 2007), which we are going to investigate in the second semester.

6 Summary and Discussion

The goal of this project is to study the impact of electron-electron interactions on the Hall conductivity of massive Dirac fermions in a gapped hexagonal lattice. In order to do so, we start with a general description of the topological and geometrical aspect of the band structure, which allows us to relate the Hall conductivity to a topological invariant of the system, the Chern number. Continuing with the basis of linear response theory, we choose to limit our approach to the study of periodic perturbations, on the grounds that well-behaved functions can be written as a superposition of periodic functions via a Fourier transform. After introducing the Lehmann representation and evaluating the transverse current response, we show that it is directly related to the Hall conductivity in the limit $\omega \to 0$, by the Kubo formula.

For the specific case of massive Dirac fermions, we calculate the Hall conductivity when the Fermi energy is inside, below or above the gap, and we obtain results consistent with Sinitsyn et al. (2006). Similar to the way this paper investigates the robustness of the Hall conductivity under the influence of disorder, we pursue into examining the effects of electron-electron interactions, by means of diagrammatic perturbation theory.

We then introduce the time-ordered correlation and the non-interacting Green's functions under the adiabatic assumption, which ensures the interaction is slowly "switched on". Evaluating the zero-th order Feynman diagram gives the same result for the Hall conductivity as in the linear response approach. However, when trying to compute a first-order Hartree-Fock correction to the correlation function, both infrared and ultraviolet divergences are encountered, which can be dealt with using various regularisation methods.

Coming back to our initial discussion of perturbation theory, we remember that our formalism is only valid for small coupling constants, while we are in fact interested in an interaction whose strength is of order unity. This means that all the orders of the power series have in fact similar weights, which justifies why it is not enough to only consider a first-order correction. In light of this, our plan for the future work is to consider instead diagrams that are "building blocks" for an infinite-order resummation, which is to be performed by means of integral equations (Giuliani, Vignale, 2005).

A Proof of the Lehmann Representation

The Lehmann representation is also known as the exact eigenstates representation since it allows us to express the response function in terms of the complete set of eigenstates of the non-interacting Hamiltonian. The expectation value of the commutator is:

$$\begin{aligned}
\langle[\hat{A}(\tau),\hat{B}(0)]\rangle_0 &= \sum_n P_n \langle\psi_n|\left(\hat{A}(\tau)\hat{B}(0)-\hat{B}(0)\hat{A}(\tau)\right)|\psi_n\rangle \\
&= \sum_n P_n \langle\psi_n|\left(e^{i\hat{H}\tau}\hat{A}\,e^{-i\hat{H}\tau}\hat{B}-\hat{B}\,e^{i\hat{H}\tau}\hat{A}\,e^{-i\hat{H}\tau}\right)|\psi_n\rangle \\
&= \sum_n P_n\, e^{i\varepsilon_n\tau}\langle\psi_n|\,\hat{A}\Big(\sum_m |\psi_m\rangle\langle\psi_m|\Big)e^{-i\hat{H}\tau}\hat{B}\,|\psi_n\rangle - \\
&\quad - \sum_n P_n\, e^{-i\,\varepsilon_n\tau}\langle\psi_n|\,\hat{B}\Big(\sum_m |\psi_m\rangle\langle\psi_m|\Big)e^{i\hat{H}\tau}\hat{A}\,|\psi_n\rangle \\
&= \sum_{n,m} P_n\Big(e^{i\omega_{nm}\tau}A_{nm}B_{mn}-e^{i\omega_{mn}\tau}B_{nm}A_{mn}\Big) \\
&= \sum_{n,m} P_n e^{i\omega_{nm}\tau}A_{nm}B_{mn}-\sum_{n,m}P_m e^{i\omega_{nm}\tau}B_{mn}A_{nm} \\
&= \sum_{n,m}(P_n-P_m)e^{i\omega_{nm}\tau}A_{nm}B_{mn}, \qquad \text{(A.1)}
\end{aligned}$$

where $A_{nm}=\langle\psi_n|\,\hat{A}\,|\psi_m\rangle$, $B_{mn}=\langle\psi_m|\,\hat{B}\,|\psi_n\rangle$ and $\omega_{nm}=\varepsilon_n-\varepsilon_m$. Relabelling $n\leftrightarrow m$ and making use of the fact that $\omega_{nm}=-\omega_{mn}$:

$$\begin{aligned}
\chi_{AB}(\omega) &= -i\int_0^\infty \langle[\hat{A}(\tau),\hat{B}(0)]\rangle_0\; e^{i(\omega+i\eta)\tau}d\tau. \\
&= -i\int_0^\infty \sum_{n,m}(P_m-P_n)e^{-i\omega_{nm}\tau}A_{mn}B_{nm}\,e^{i(\omega+i\eta)\tau}d\tau \\
&= -\cancel{i}\sum_{n,m}\frac{P_m-P_n}{\cancel{i}(\omega-\omega_{nm}+i\eta)}\left[e^{i(\omega-\omega_{nm}+i\eta)\tau}\right]_0^\infty A_{mn}B_{nm} \\
&= -\cancel{i}\sum_{n,m}\frac{P_m-P_n}{\cancel{i}(\omega-\omega_{nm}+i\eta)}\left[e^{i(\omega-\omega_{nm})\tau}\,e^{-\eta\tau}\right]_0^\infty A_{mn}B_{nm} \\
&= \sum_{n,m}\frac{P_m-P_n}{\omega-\omega_{nm}+i\eta}A_{mn}B_{nm}. \qquad \text{(A.2)}
\end{aligned}$$

At zero temperature only the ground state is occupied, so we can take $n=0$ in the previous equations since all the other probabilities vanish. This gives an alternative form for the response function, which will prove useful for our discussion in Appendix C:

$$\chi_{AB}(\omega)=\sum_n\frac{A_{0n}B_{n0}}{\omega-\omega_{n0}+i\eta}-\frac{A_{n0}B_{0n}}{\omega+\omega_{n0}+i\eta}, \qquad \text{(A.3)}$$

where we switched $\omega_{0n}\to-\omega_{n0}$ since $\omega_{n0}=\varepsilon_n-\varepsilon_0>0$. Note that we relabelled $m\to n$.

B Feynman Rules

Step 1: Draw all the topologically distinct connected diagrams formed from two external vertices $j^x_{\boldsymbol{q}}$ and $j^y_{-\boldsymbol{q}}$, n oriented interaction line, and $2n+2$ oriented solid lines.

Step 2: Associate a frequency ε_i and a wave vector $\boldsymbol{k}_i$ to each solid and interaction line. Assign an outgoing frequency ω, together with a wave vector $\boldsymbol{q}$ at the vertex $j^x_{\boldsymbol{q}}$ and an incoming frequency $-\omega$, together with a wave vector $-\boldsymbol{q}$ at the vertex $j^y_{-\boldsymbol{q}}$. Now conserve the frequency and the wave vector at every vertex.

Step 3: Associate a non-interacting Green's function $G^{(0)}_{\lambda_i}(\boldsymbol{k}_i, \varepsilon_i)$ to each solid line.

Step 4: Associate a factor $v_{\boldsymbol{p}}$ to the interaction line, where $\boldsymbol{p}$ is the momentum transfer in the scattering process.

Step 5: Associate the matrix element $\langle \psi^{\lambda_2}_{\boldsymbol{k}_2} | \sigma_x | \psi^{\lambda_1}_{\boldsymbol{k}_1} \rangle$ to the external vertex $j^x_{\boldsymbol{q}}$ and the matrix element $\langle \psi^{\lambda_4}_{\boldsymbol{k}_4} | \sigma_y | \psi^{\lambda_3}_{\boldsymbol{k}_3} \rangle$ to the vertex $j^y_{\boldsymbol{q}}$, where $\boldsymbol{k}_1$, $\boldsymbol{k}_2$ and $\boldsymbol{k}_3$, $\boldsymbol{k}_4$ are the wave vectors of the outgoing and incoming lines at the two vertices, respectively.

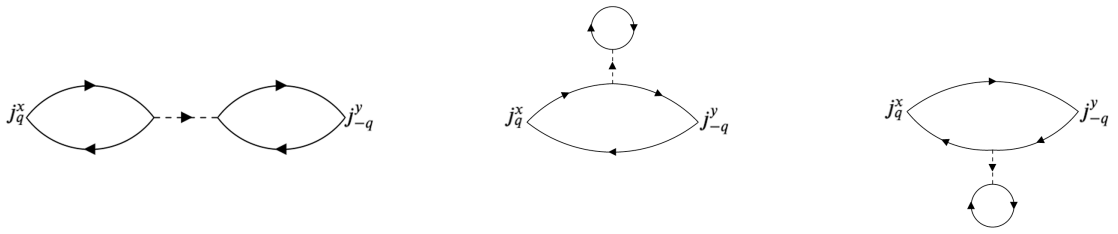
Step 6: Associate the matrix elements $\langle \psi^{\lambda_3}_{\boldsymbol{k}_3} | \psi^{\lambda_2}_{\boldsymbol{k}_2} \rangle$ and $\langle \psi^{\lambda_1}_{\boldsymbol{k}_1} | \psi^{\lambda_4}_{\boldsymbol{k}_4} \rangle$ to the internal vertices.

Step 7: Integrate over the $n+1$ internal wave vectors and $n+1$ frequencies, each with a measure $\int d\boldsymbol{k}_i/(2\pi)^{\mathrm{D}}$ and $\int d\varepsilon_i/2\pi$. Sum over all the spin states λ_i.

Step 8: Multiply by a factor $i^{n+1}(-1)^L$, where L is the number of fermion loops.

Step 9: Interpret the integral $\int d\varepsilon_i/2\pi\, G^{(0)}_{\lambda_i}(\boldsymbol{k}_i, \varepsilon_i)$ as $\lim_{\eta\to 0^+} \int d\varepsilon_i/2\pi\, G^{(0)}_{\lambda_i}(\boldsymbol{k}_i, \varepsilon_i)\, e^{i\eta\omega}$.

For the zero-th order, the free variables are $\boldsymbol{k}$ and ε. For the first order, only the two internal vertices give rise to constraints when conserving the frequency and the wave vector, as the external vertices conserve them automatically when assigning them $\pm\omega$ and $\pm\boldsymbol{q}$. The two pairs of delta functions eliminate 4 "dummy" variables - 2 frequencies and 2 wave vectors - such that the only free variables we are integrating over are: ε, ε', $\boldsymbol{k}$ and $\boldsymbol{k}'$.



(a) (b) (c)

Figure B.1: First order Feynman diagrams which we neglect in our calculation.

C Time-Ordered Correlation Function

In accordance with the adiabatic assumption, the time-ordered correlation is given by:

$$\chi^T_{AB}(\omega) = \lim_{\eta\to 0^+} \int_{-\infty}^{\infty} \chi^T_{AB}(t) e^{i\omega t - \eta|t|} dt. \tag{C.1}$$

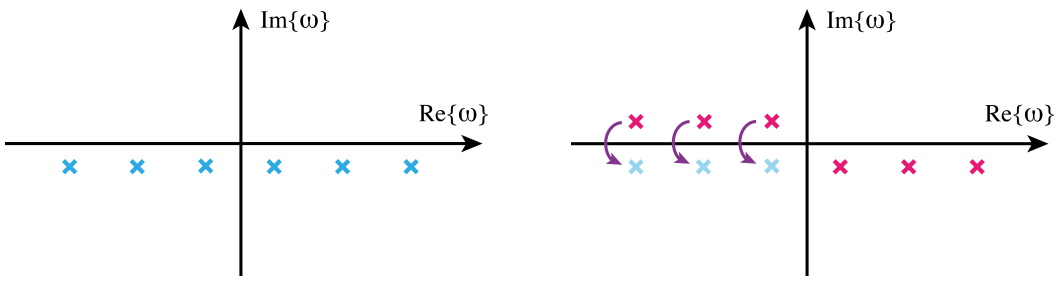
Since we need to evaluate the expectation value of the time-ordered product of $\hat{A}(t)\hat{B}(0)$ for all times, we can split the time integral by the sign of t:

$$\begin{aligned}
\chi^T_{AB}(\omega) &= -i \int_{-\infty}^{0} \langle \hat{B}(0)\hat{A}(t)\rangle_0 \, e^{i\omega t} \, e^{+\eta t} dt - i \int_0^{\infty} \langle \hat{A}(t)\hat{B}(0)\rangle_0 \, e^{i\omega t} \, e^{-\eta t} dt \\
&= -i \sum_{n,m} P_n \int_{-\infty}^{0} e^{i(\omega - \omega_{nm} - i\eta)t} B_{nm} A_{mn} dt - i \sum_{n,m} P_n \int_0^{\infty} e^{i(\omega + \omega_{nm} + i\eta)t} A_{nm} B_{mn} dt \\
&= -i \sum_{n,m} P_n B_{nm} A_{mn} \left[e^{i(\omega - \omega_{nm} - i\eta)t} \right]_{-\infty}^{0} - i \sum_{n,m} P_n A_{nm} B_{mn} \left[e^{i(\omega + \omega_{nm} + i\eta)t} \right]_0^{\infty} \\
&= -\cancel{i} \sum_{n,m} \frac{P_n B_{nm} A_{mn}}{\cancel{i}(\omega - \omega_{nm} - i\eta)} + \cancel{i} \sum_{n,m} \frac{P_n A_{nm} B_{mn}}{\cancel{i}(\omega + \omega_{nm} + i\eta)} \\
&= \sum_{n,m} P_n \left(\frac{A_{nm} B_{mn}}{\omega + \omega_{nm} + i\eta} - \frac{B_{nm} A_{mn}}{\omega - \omega_{nm} - i\eta} \right).
\end{aligned} \tag{C.2}$$

At zero temperature, we again take $n = 0$ and get the time-ordered correlation function:

$$\chi^T_{AB}(\omega) = \sum_n \frac{A_{0n} B_{n0}}{\omega - \omega_{n0} + i\eta} - \frac{B_{0n} A_{n0}}{\omega + \omega_{n0} - i\eta}. \tag{C.3}$$

Comparing this expression to the Lehmann representation of the response function given in (A.3), we can see that the only difference between the two is the position of the poles: while the response function has all poles in the lower half-plane, the time-ordered correlation has poles in the lower half-plane only for positive frequencies, while for negative ones the poles are in the upper half-plane. This is shown in Figure C.1.



(a) Lehmann representation (b) Correlation function

Figure C.1: The poles of the response and time-ordered correlation functions.

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